

**PROGRAMME: BCA (Bachelor of Computer Applications)
SEMESTER – III**

Teaching-Learning & Evaluation Plan

Course Information:

Course Code: 23BCA3C01	Course Title: Software Engineering	
Credits Units: 03	Contact Hours: 45-45	L-T-P-E: 3-0-0-3
IA: UE Weightage - 50: 50	Pass Marks (IA & UE) – 40 (UE-Min. 18)	Aggregate Pass Marks: 40

UE Question Paper Marks: **50** Special Examination Fees: **NA**

Pre-requisite (if any):

1. Student should have completed a basic course on Computer and Programming

Course Facilitator (s): Dr. Deepak Mehta, Ms. Saumya Patel, Mr. Diwakar Vasuman, Mr. Saravanan, Dr. Ananta Ojha, SOUMEN CHAKRABORTY

PO	Graduate Attribute	Description
PO1	Computational Knowledge	Understand and apply mathematical, computing, and domain knowledge for conceptualizing computer-based solutions for real-world problems.
PO2	Problem Analysis	Identify, critically analyze, and formulate complex computing problems using the knowledge of computer science and information technology.
PO3	Design and Development of Solutions	Design and develop computer-based systems and components for solving complex real-world problems using appropriate computing technologies.
PO4	Conduct Investigations of Complex Computing Problems	Design and conduct experiments, analyze and interpret data, and provide valid conclusions.
PO5	Modern Tool Usage	Recognize and use modern computing tools, and techniques necessary for innovative computer-based solutions.
PO6	Project Management	Demonstrate engineering and management principles and practices to manage projects in multidisciplinary environments.

PO7	Innovation and Entrepreneurship	Identify opportunities, use innovative ideas, and entrepreneurship mindset to create value and wealth for the betterment of self and society.
PO8	Communication Skills	Communicate effectively in the workplace, collaborate, and contribute to achieve common goals.
PO9	Individual and Team Work	Work as a member or leader in diverse teams in a multidisciplinary environment.
PO10	Professional Ethics	Exhibit the professional attitude and ethics needed for a successful career in industry and academia.
PO11	Societal & Environmental Concerns	Recognize the social and environmental impacts of computer technology and demonstrate professional practices needed for sustainable development.
PO12	Life-long Learning	Recognize the need for and develop the ability to engage in continuous learning for personal and organizational growth.

Course Objectives (COB's)

COB1: To understand the concepts of Software Engineering and SDLC life cycle phases.

COB2: To examine the software requirements and develop skills in software designing to solve real life problems.

COB3: To learn and apply software testing techniques to test real life software applications.

Course Outcomes (CO's)

At the end of the course, students will be able to

CO1: Describe the software product, process and SDLC models.

CO2: Identify software requirements and prepare SRS document.

CO3: Construct various analysis models for a software system.

CO4: Construct different design models for a software system.

CO5: Implement various testing strategies and techniques to test software system.

S. No.	Course Outcome	Skill's	POs	Bloom's Taxonomy Level
1.	CO 1	Understanding	PO1, PO7, PO8	L2
2.	CO 2	Identify	PO1, PO2, PO4, PO7	L2
3.	CO 3	Create	PO3, P PO4, PO5, PO6, PO9, PO12	L6
4.	CO 4	Create	P01, P02, P03, P05, PO9, P010, PO11, PO12	L6
5.	CO 5		P01, PO3, P04, P05, PO6, PO9, P010	

		Implement		L3
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CO-PO/PSO Mapping: (3-Strong Correlation 2- Medium Correlation 1- Low Correlation)

Course Outcomes	Blooms Taxonomy Level	Programme Outcomes (PO)												Programme Specific Outcomes (PSO)			
		PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PSO 1 L2	PSO 2 L3	PSO 3 L4	PSO 4 L6
CO01	L2	1	-	-	-	-	-	2	3	-	-	-	-	1	1	3	2
CO02	L2	1	3	-	3	-	-	2	-	-	-	-	-	1	3	1	2
CO03	L6	-	-	1	2	2	1	-	-	2	-	-	3	3	1	2	1
CO04	L3	2	1	3	-	2	-	-	-	2	3	2	3	3	1	3	2
CO05	L2	3	-	3	2	2	-	-	-	2	1	3	2	3	3	2	3
Total		7	4	7	7	6	1	4	3	6	4	5	8	11	9	11	10
Average		1.4	0.8	1.4	1.4	1.2	0.2	0.8	0.6	1.2	0.8	1	1.6	2.2	1.8	2.2	2

Course Contents:

Module	Details	Contact Hours
I	Software Product and Process: Defining Software, Categories of Software, Legacy Software, Changing Nature of Software, Defining Software Engineering, Software Process Framework, SDLC, Prescriptive Process Models: Waterfall, V-Model, Incremental Model, Prototyping, Spiral Model; Agile Development: Agile Process, Extreme Programming, Scrum; Characteristics of a Software Engineer, Software Team	9

II	Software Requirements: Understanding Requirements: What are Software Requirements? Characteristics of Good Requirements, Functional and Non-Functional Requirements, Requirements Engineering Process: Feasibility Studies, Elicitation, Specification, Validation; Software Requirements Specification (SRS) document, IEEE Guidelines	9
III	Requirements Modelling- Scenario-Based Modeling: Use-Case Model, Use Case Specification, Activity Diagram; Class-Based Modeling: Identifying Analysis Classes, CRC Model, Association and Dependency, Class Diagram; Behavior Modeling: State Diagram, Sequence Diagram, and Collaboration Diagram; Data Modeling: E-R Diagram, Mapping E-R diagram to Relational Model.	9
IV	Design Models: Fundamental Design Concepts, Software Architecture: importance of software architecture, architectural styles and patterns; Component-level Design: What is a component, Basic Design Principles, Design guidelines, Cohesion, Coupling, Component Diagram, Deployment Diagram; User Interface Design: The Golden Rules, interface design evaluation cycle. Introduction to Pattern Based Design: MVC pattern, Data Access Object pattern, WebApp Design: Design quality, Goals, Design Pyramid; Mobile App Design: Challenges, Design Best Practices.	9
V	Software Testing: Verification and Validation, Testing Strategies, Unit testing, and Integration Testing, Validation testing, System testing, the Art of Debugging, Black-box, and White-box testing, Basis path testing, Control structure testing, Graph-Based testing, Equivalence partitioning, Boundary value analysis, Object-Oriented testing, Scenario-Based Testing, Random testing, Partition testing, Inter-class testing; Introduction to automated testing tools.	9

Text Books (TB):

TB-1	1. Roger S. Pressman, “Software Engineering – A Practitioner’s Approach”, Eighth Edition, McGraw-Hill International Edition, 2015
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Reference Books (RB):

RB-1	Ian Sommerville, “Software engineering”, Tenth Edition, Pearson Education, 2015
RB-2	Waman S. Jawadekar Software Engineering: Principles and Practice, Tata McGraw-Hill Education, 2008
RB-3	Rajib Mall, “Fundamentals of Software Engineering”, PHI, 5th Edition, 2018

RB-4	Deepak Jain, “Software Engineering, Principles and Practices”, Oxford University Press, 2008.
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Relevant Website (RW):

RW-1	https://nptel.ac.in/courses/1061010612
RW-2	https://www.youtube.com/watch?v=tZreaH_FyMs
RW-3	https://www.udemy.com/course/the-complete-software-engineering-from-basics-to-advanced/
RW-4	https://www.youtube.com/watch?v=LxEmGNgqYJA
RW-5	http://iansommerville.com/software-engineering-book/slides/

Video Links (VL):

VL-1	Basics of SE: https://www.youtube.com/watch?v=sB2iQSvrcG0
VL-2	S/W Engineering Paradigm: https://www.youtube.com/watch?v=UTOsHHonS5g
VL-3	Life Cycle Models: https://www.youtube.com/watch?v=l9oQwHOQB18
VL-4	Functional and Non-Functional Requirements: https://www.youtube.com/watch?v=j4WITZFLkUM
VL-5	Software Document: https://www.youtube.com/watch?v=4fYy4l3sSKU
VL-6	Requirement Engineering Process – Feasibility Studies: https://www.youtube.com/watch?v=Z9mKgHbfPpE
VL-7	Analysis Concepts - Design Process And Concepts: https://www.youtube.com/watch?v=sB2iQSvrcG0
VL-8	Modular Design – Design Heuristic – Architectural Design: https://www.youtube.com/watch?v=IPIP2R7l-Nc
VL-9	Collaboration Diagram: https://www.youtube.com/watch?v=WTiZM3LVVJg
VL-10	E-R Diagrams: https://www.youtube.com/watch?v=5nGC4fyFPes
VL-11	Component level design: https://www.youtube.com/watch?v=Nz4JTEKBVNg
VL-12	Coupling and cohesion: https://www.youtube.com/watch?v=TCMGU7-Ir_k
VL-13	User Interface Design: https://www.youtube.com/watch?v=_K06Dni-RE4
VL-14	MVC controller https://www.youtube.com/watch?v=DUg2SWWK18I
VL-15	Taxonomy of Software Testing – Types of S/W Test: https://www.youtube.com/watch?v=cl6pNHGHQEQ
VL-16	Black Box Testing: https://www.youtube.com/watch?v=dQ2EQnkyOTMeo

VL-17	Unit Testing: https://www.linkedin.com/learning/software-testing-foundations-test-management/starting-out-with-test-management?u=92695330
VL-18	Software testing: https://www.youtube.com/watch?v=CjqAct2ySv0
VL-19	Basis Path testing: https://www.youtube.com/watch?v=J_cAqU3mzyM
VL-20	Automation testing: https://www.youtube.com/watch?v=QtZ4yV49RtA

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	Software Product & Process	CO1	Software Development Life Cycle (SDLC) https://www.linkedin.com/learning/software-development-life-cycle-sdlc/processes-for-software-projects?u=92695330	1 h 34m
MC-2	LinkedIn Learning	Agile Foundations	CO1	Agile Foundations: https://www.linkedin.com/learning/agile-foundations?trk=learning-topics_trending-courses_related-content-card&upsellOrderOrigin=default_guest_learning	1 h 35m
MC-3	LinkedIn Learning	Scrum: The Basics	CO1	Scrum: The Basics: https://www.linkedin.com/learning/scrum-the-basics?trk=learning-topics_trending-courses_related-content-card&upsellOrderOrigin=default_guest_learning	1h
MC-4	LinkedIn Learning	Software Requirements	CO2	Requirements Elicitation and Analysis https://www.linkedin.com/learning/requirements-elicitation-and-analysis/start-defining-and-discovering?u=92695330	1h 22m
MC-5	LinkedIn Learning	Software Requirements	CO2	Software Design: Developing Effective Requirements https://www.linkedin.com/learning/software-design-developing-effective-requirements/what-you-should-know?u=92695330	1h 45m
MC-6	LinkedIn Learning	Analysis & Design	CO3	Programming Foundations: Object-Oriented Design https://www.linkedin.com/learning/programming-foundations-object-oriented-design-3/learn-object-oriented-design-principles?u=92695330	2 hr 40 m
MC-7	LinkedIn Learning	Testing	CO4	Test Automation Foundations https://www.linkedin.com/learning/test-automation-foundations/build-a-foundation-of-test-automation?contextUrn=urn%3Ali%3AlyndaLearningPath%3A57f7e27c3dd559e018dfe994&u=92695330	2 hr 9m

MC-8	LinkedIn Learning	Software Architecture Foundations	CO4	Software Architecture Foundations https://www.linkedin.com/learning/software-architecture-foundations?trk=learning-topics_trending-courses_related-content-card&upsellOrderOrigin=default_guest_learning	1hr 36m
MC-9	LinkedIn Learning	Software Design: Modeling with UML	CO4	Software Design: Modeling with UML https://www.linkedin.com/learning/software-design-modeling-with-uml	1h 41m
Total MOOC Course integration with Certification					15 hours

Session-wise Planning:

Module 1				
Session	Topic	Readings & References	CO	Mode of Delivery
1.	Defining Software, Categories of Software,	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=sB2iQSvrcG0	CO1	PPT Presentation & Classroom interaction
2.	Legacy Software, Changing Nature of Software	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=UTOsHHonS5g	CO1	PPT Presentation & Classroom Interaction
3.	Defining Software Engineering, Software Process Framework, SDLC	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015	CO1	PPT Presentation & Classroom Interaction

		2. Video Link: https://www.youtube.com/watch?v=I9oQwHOQBI8 https://www.slideshare.net/amralsayedm/se18lec-02software-life-cycle-model SDLC Models: https://www.slideshare.net/amralsayedm/se18lec-02software-life-cycle-model		
4.	Prescriptive Process Models: Waterfall	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 Video Link: https://www.youtube.com/watch?v=I9oQwHOQBI8 https://www.slideshare.net/amralsayedm/se18lec-02software-life-cycle-model	CO1	PPT Presentation & Classroom interaction
5.	V-Model, Incremental Model	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. https://www.slideshare.net/amralsayedm/se18lec-02software-life-cycle-model	CO1	PPT Presentation & Classroom interaction
6.	Prototyping, Spiral Model;	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. https://www.slideshare.net/amralsayedm/se18lec-02software-life-cycle-model	CO1	PPT Presentation & Classroom interaction
7.	Agile Development: Agile Process,	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. https://www.youtube.com/watch?v=I9oQwHOQBI8	CO1	PPT Presentation & Classroom interaction
8.	Extreme Programming, Scrum;	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”,	CO1	PPT Presentation & Classroom interaction

		Eighth Edition, McGraw-Hill, 2015 2.https://www.youtube.com/watch?v=l9oQwHOQBI8		
9.	Characteristics of a Software Engineer, Software Team	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2.https://www.youtube.com/watch?v=l9oQwHOQBI8	CO1	PPT Presentation & Classroom interaction
<u>Module 2</u>				
10.	What are Software Requirements?	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=j4WITZFLkUM	CO2	PPT Presentation & Classroom interaction
11.	Characteristics of Good Requirements	1. Ian Sommerville, “Software engineering”, Tenth Edition, Pearson Education, 2015 2. Video Link: https://www.youtube.com/watch?v=4fYy4l3sSKU	CO2	PPT Presentation & Classroom interaction
12.	Functional and Non-Functional Requirements	1. Ian Sommerville, “Software engineering”, Tenth Edition, Pearson Education, 2015 2. Video Link: https://www.youtube.com/watch?v=Z9mKgHbfPpE	CO2	PPT Presentation & Classroom interaction
13.	Requirements Engineering Process	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO2	PPT Presentation & Classroom interaction
14.	Feasibility Studies	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO2	PPT Presentation & Classroom interaction
15.	Elicitation	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”,	CO2	PPT Presentation & Classroom interaction

		Eighth Edition, McGraw-Hill, 2015		
16.	Specification	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015	CO2	PPT Presentation & Classroom interaction
17.	Validation	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=sB2iQSvrcG0	CO2	PPT Presentation & Classroom interaction
18.	Software Requirements Specification (SRS) document, IEEE Guidelines	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=sB2iQSvrcG0	CO2	PPT Presentation & Classroom interaction
Module 3				
19.	Scenario-Based Modeling	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=IPI2R7l-Nc	CO3	PPT Presentation & Classroom interaction
20.	Use-Case Model, Use Case Specification,	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015	CO3	PPT Presentation & Classroom interaction
21.	Activity Diagram; Class-Based Modeling	1. Waman S. Jawadekar Software Engineering: Principles and Practice, Tata McGraw-Hill Education, 2008	CO3	PPT Presentation & Classroom interaction
22.	Identifying Analysis Classes, CRC Model,	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015	CO3	PPT Presentation & Classroom interaction

23.	Association and Dependency	1. Waman S. Jawadekar Software Engineering: Principles and Practice, Tata McGraw-Hill Education, 2008	CO3	PPT Presentation & Classroom interaction
24.	Class Diagram; Behavior Modeling	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=cl6pNHGHQEQ	CO3	PPT Presentation & Classroom interaction
25.	State Diagram, Sequence Diagram	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video links: https://www.linkedin.com/learning/software-testing-foundations-test-management/starting-out-with-test-management?u=92695330	CO3	PPT Presentation & Classroom interaction
26.	Collaboration Diagram; Data Modeling	1. Waman S. Jawadekar Software Engineering: Principles and Practice, Tata McGraw-Hill Education, 2008 2. Video Link: https://www.youtube.com/watch?v=WTiZM3LVVJg	CO3	PPT Presentation & Classroom interaction
27.	E-R Diagram, Mapping E-R diagram to Relational Model	1. Deepak Jain, “Software Engineering, Principles and Practices”, Oxford University Press, 2008. 2. https://www.youtube.com/watch?v=5nGC4fyFPes	CO3	PPT Presentation & Classroom interaction
<u>Module 4</u>				
28.	Fundamental Design Concepts, Software Architecture	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video link: https://www.youtube.com/watch?v=dQ2EQnkyOTMeo	CO4	PPT Presentation & Classroom interaction

29.	importance of software architecture, architectural styles and patterns	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video links https://www.youtube.com/watch?v=dQ2EQnkyOTMeo	CO4	PPT Presentation & Classroom interaction
30.	Component-level Design: What is a component, Basic Design Principles	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video link: https://www.youtube.com/watch?v=Nz4JTEKBNg	CO4	PPT Presentation & Classroom interaction
31.	Design guidelines, Cohesion, Coupling	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video link: https://www.youtube.com/watch?v=TCMGU7-Ir_k	CO5	PPT Presentation & Classroom interaction
32.	Component Diagram, Deployment Diagram	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.linkedin.com/pulse/software-project-management-madhur-modi/	CO4	PPT Presentation & Classroom interaction
33.	User Interface Design: The Golden Rules, interface design evaluation cycle	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015 2. Video link: https://www.youtube.com/watch?v=_K06Dni-RE4	CO4	PPT Presentation & Classroom interaction
34.	Introduction to Pattern Based Design: MVC pattern	1. Roger S. Pressman, "Software Engineering – A practitioner's Approach", Eighth Edition, McGraw-Hill, 2015	CO4	PPT Presentation & Classroom interaction

		2. Video link: https://www.youtube.com/watch?v=DUg2SWWK18I		
35.	Data Access Object pattern, WebApp Design: Design quality, Goals	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video link: https://www.youtube.com/watch?v=DUg2SWWK18I	CO4	PPT Presentation & Classroom interaction
36.	Design Pyramid; Mobile App Design: Challenges, Design Best Practices	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO4	PPT Presentation & Classroom interaction
<u>Module 5</u>				
37.	Verification and Validation, the Art of Debugging	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=CjqAct2ySv0	CO5	PPT Presentation & Classroom interaction
38.	Black-box and White-box testing, Equivalence partitioning	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=CjqAct2ySv0	CO5	PPT Presentation & Classroom interaction
39.	Boundary value analysis, Testing Strategies	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=CjqAct2ySv0	CO5	PPT Presentation & Classroom interaction
40.	Unit testing and Integration Testing	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO5	PPT Presentation & Classroom interaction

		2.Video Link: https://www.youtube.com/watch?v=CjqAct2ySv0		
41.	Validation testing, System testing	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO5	PPT Presentation & Classroom interaction
42.	Basis path testing, Control structure testing,	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2.Video Link: https://www.youtube.com/watch?v=J_cAqU3mzyM	CO5	PPT Presentation & Classroom interaction
43.	Graph-Based testing, Object-Oriented testing,	1.Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 2. Video Link: https://www.youtube.com/watch?v=J_cAqU3mzyM	CO5	PPT Presentation & Classroom interaction
44.	Scenario-Based Testing, Radom testing, Partition testing,	1. Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015	CO5	PPT Presentation & Classroom interaction
45.	Inter-class testing; Introduction to automated testing tools.	Roger S. Pressman, “Software Engineering – A practitioner’s Approach”, Eighth Edition, McGraw-Hill, 2015 Video Link https://www.youtube.com/watch?v=QtZ4yV49RtA	CO5	PPT Presentation & Classroom Interaction

Assessment Scheme: CA: UE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency/Stages	Weightage (%)	CO
1.	Skill Enhancement Activity-1 – LinkedIn Certification on SE Concepts	Formative	1	15	CO1, CO2, CO3, CO4, CO5
2.	Content Improvement Activity-2 – Mini		1/ 2	15	CO2, CO3, CO5

	Project on SE Concepts				
3.	Class Participation	Formative	Continuous	5	-CO1, CO2, CO3, CO4, CO5
4	Internal Assessment Test I	Formative	1	7.5	CO1, CO2
	Internal Assessment Test II	Formative	1	7.5	CO3, CO4, CO5
5.	End Semester Examination	Summative	1	50	CO1, CO2, CO3, CO4, CO5
Total				100	

Assessment Sheet with Rubrics for Grading & Evaluation

Activity 1: Online Certification

Criteria	Poor Below Expectation	Fair Needs Improvement	Good Meets Expectation	Excellent Exceeds Expectation	Weightage
Certificate Completion	(5-6 marks) Completed less than 15 hrs in 1-2 days	(7-8 marks) Completed 15 hrs or less in 2-3 days	(9 marks) Completed required 15 hrs in less 5-6 days	(10 marks) Completed required 15 hrs in 7-8 days	10
Report Submission	(2 marks) Poor Clarity in lessons learned; poorly organised report	(3 marks) Moderate clarity in lessons learned & report	(4 marks) Good Clarity in lessons learned & organised report	(5 marks) Very Good Clarity in lessons learned & well organised report	5
Oral Presentation	(2 marks) Poor language, voice, & Content	(3 marks) Moderate language, voice, & Content	(4 marks) Good language, voice, & Content	(5 marks) Very Good language, voice, & Content	5
Question-Answer	(2 marks) Answers less than 3 questions correctly	(3 marks) Answers 3 questions correctly	(4 marks) Answers 4 questions correctly	(5 marks) Answers all 5 questions correctly	5

Activity 2 – Mini Project (SRS document and System Design)

Criteria	Poor Below Expectation	Fair Needs Improvement	Good Meets Expectation	Excellent Exceeds Expectation	Weightage
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Report Submission	(8 marks) Poor: Poor selection of topic, No Clarity on literature survey and design & plagiarism	(10-11 marks) Moderate: Selected a topic through improper knowledge, No much Clarity in literature survey, design & No plagiarism	(12-13 marks) Good Report: Selected a topic with partial knowledge and did considerable literature survey, design with standard references and No plagiarism	(14-15 marks) Very Good Report: Selected a latest topic with Extensive literature survey, design with standard references and No plagiarism	15
Oral Presentation	(2 marks) Poor language, voice, & Content	(3 marks) Moderate language, voice, & Content	(4 marks) Good language, voice, & Content	(5 marks) Very Good language, voice, & Content	5
Question-Answer	(2 marks) Answers less than 3 questions correctly	(3 marks) Answers 3 questions correctly	(4 marks) Answers 4 questions correctly	(5 marks) Answers all 5 questions correctly	5